

Short Answers (2 Marks).

Q1. What is mobile application development?

Ans: Mobile application development is the process of designing, coding, testing, and deploying software applications that run on mobile devices such as smartphones and tablets.

Q2. List any two popular mobile platforms and their key differences.

Ans: Android is an open-source platform used by multiple manufacturers, while iOS is a closed-source platform used only on Apple devices.

Q3. What do you mean by the Android ecosystem?

Ans: The Android ecosystem includes the Android OS, Google Play services, apps, developer tools, hardware manufacturers, and services that together support Android usage and development.

Q4. What is Android Studio?

Ans: Android Studio is the official Integrated Development Environment (IDE) by Google used to build, test, and debug Android applications.

Q5. Name two features of the Android development environment. Ans:

1. Built-in emulator for testing.
 2. Smart code editor with autosuggestions.
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Q6. What are UI components in Android? Give two examples.

Ans: UI components are visual elements that users interact with. Examples: Button, TextView.

Q7. What is a layout in Android?

Ans: A layout defines the arrangement of UI elements on the screen such as LinearLayout or Constraint Layout.

Q8. Why is it important to support multiple screen sizes in mobile apps? Ans:

Devices have different sizes and resolutions, so supporting multiple screens ensures a consistent user experience.

Q9. What is the purpose of XML in Android UI development?

Ans: XML is used to design the user interface separately from application logic, which helps manage and update UI easily.

Q10. Name any three commonly used UI components in Android. Ans: TextView, EditText, RecyclerView.

Q11. What are Android styles and themes?

Ans: Styles define the look of individual UI components, while themes apply a unified design across the entire app.

Q12. Define Android Studio. Name any two major components it provides. Ans: Android Studio is Google's IDE for Android development. Components: Layout Editor, Gradle Build System.

Q13. What are custom views in Android?

Ans: Custom views are user-defined components created to add special behaviors and appearance not available in built-in views.

Q14. What is the use of RecyclerView in Android?

Ans: RecyclerView is used to efficiently display large lists or grids by reusing views to improve scrolling performance.

Q15. What is SQLite in Android, and why is it used?

Ans: SQLite is a lightweight relational database used to store structured data locally inside an application.

Q16. Define Shared Preferences in Android.

Ans: Shared Preferences store small key-value pair data such as user settings or login information.

Q17. List properties of TextView in Android. Explain any two.

Ans: Properties include text, textSize, textColor, gravity, and maxLines. textSize controls font size; gravity sets alignment of text.

Q18. Differentiate between Internal Storage and External Storage in Android. Ans: Internal storage is private and secure for the app, while external storage is shared among apps and requires permissions.

Q19. What are RESTful web services in Android?

Ans: RESTful web services are APIs that use HTTP methods to transfer data between client and server efficiently.

Q20. List three key classes used for SQLite database handling in Android. Ans: SQLiteOpenHelper, SQLiteDatabase, Cursor.

Q21. What is MediaStore in Android?

Ans: MediaStore is a system-level content provider that manages multimedia files like images, audio, and video.

Q22. Name two APIs used for playing audio in Android. Ans: MediaPlayer and ExoPlayer.

Q23. What is the purpose of MediaScanner?

Ans: MediaScanner detects new media files in storage and updates the MediaStore database.

Q24. What is RESTful API? List commonly used HTTP methods.

Ans: A RESTful API allows communication between client and server using HTTP. Methods: GET, POST, PUT, DELETE.

Q25. Define AsyncTask in Android.

Ans: AsyncTask performs background tasks without blocking the UI thread and updates the UI after completion. (Now deprecated)

Q26. Which intent is used for photo capture in Android?

Ans: MediaStore.ACTION_IMAGE_CAPTURE

Q27. What is an HTTP request in mobile app development?

Ans: It is a network communication request used to send or receive data from a server over the internet.

Q28. List four common HTTP request methods. Ans: GET, POST, PUT, DELETE.

Q29. What is GPS in Android applications?

Ans: GPS provides the device's real-time geographical location such as latitude and longitude.

Q30. Mention two Android classes used for location services. Ans:

LocationManager, FusedLocationProviderClient.

Q31. What is unit testing in Android?

Ans: Unit testing checks individual components or functions of an app to ensure they work correctly.

Q32. What is Android Runtime (ART)? Mention its role.

Ans: ART is the runtime environment that executes app bytecode and handles memory management and garbage collection.

Q33. Name two debugging tools in Android Studio. Ans: Logcat and Android Profiler.

Long Answers (5 Marks) Q1. Explain the Android ecosystem and its main components.

Ans:

The Android ecosystem is a complete environment that supports the functioning and development of Android devices and apps. It includes multiple components working together:

- **Android Operating System:** Provides core features, security, UI, and hardware access.
- **Application Framework:** Offers APIs for app development like Activity Manager and Content Providers.
- **Android Runtime (ART):** Executes application bytecode and manages memory.
- **Libraries:** Provide essential functionalities such as graphics, media playback, and database support.
- **Linux Kernel:** Manages hardware communication and security layers.
- **Google Play Services & Play Store:** Enable app distribution, updates, authentication, and cloud services.

All these components together form a flexible and scalable mobile software ecosystem.

Q2. Describe the steps involved in creating a simple Android app using Android Studio.

Ans:

To create a simple app in Android Studio, follow these steps:

1. **Start a New Project** – Select a template such as Empty Activity and enter project details like name and package.
2. **Design UI** – Create screen layouts using XML or the visual drag-and-drop Layout Editor.
3. **Write Code Logic** – Implement functionality in Kotlin/Java files such as MainActivity.
4. **Manage Resources** – Add images, strings, and colors in resource folders.
5. **Declare Permissions** – If required, add permissions in AndroidManifest.xml.
6. **Run App** – Test using a physical device or Android Emulator.
7. **Debug & Fix Errors** – Use Logcat and Profiler for debugging.
8. **Build & Generate APK** – Finally deploy or share the application.

Q3. Discuss the role of user input handling in Android. How does the system detect touch gestures?

Ans:

User input handling allows apps to interact with users through touch, typing, gestures, and motion. Event listeners like `onClickListener` help detect user actions on UI elements.

Android detects gestures using **MotionEvent** which tracks touch actions such as:

- **ACTION_DOWN** (finger touches screen)
- **ACTION_MOVE** (finger moves on screen)
- **ACTION_UP** (finger leaves screen)

For advanced gestures, **GestureDetector** and **ScaleGestureDetector** are used to detect swipe, pinch-zoom, double-tap, etc. Android continuously monitors touch coordinates and velocity, then interprets them as gestures. This ensures smooth and responsive interaction in apps.

Q4. Explain the different types of layouts in Android with suitable examples.

Ans:

A layout arranges UI elements on the screen. Common types:

- **LinearLayout**: Places views in vertical or horizontal order (e.g., login form UI).
 - **RelativeLayout**: Positions views relative to each other or parent edges.
 - **ConstraintLayout**: Provides flexible alignment using constraints for all screen sizes.
 - **FrameLayout**: Stacks views on top of each other (e.g., overlay images).
 - **GridLayout**: Arranges components in rows and columns (e.g., gallery UI).
- Choosing the right layout improves performance and design flexibility.

Q5. Create a simple RecyclerView example that displays a list of names using Adapter and ViewHolder.

Ans:

RecyclerView is used to show large lists efficiently. Steps:

1. Add RecyclerView in XML layout.

2. Create an item layout for single row (TextView for name).
3. Create a **Model List** (array of names).
4. Create an **Adapter** class extending RecyclerView.Adapter that binds names to ViewHolder.
5. Create **ViewHolder** to store view references for reuse.
6. Set **LayoutManager** (e.g., LinearLayoutManager) and attach adapter to RecyclerView in Activity.

This reduces memory usage and gives smooth scrolling.

Q6. What are the challenges in designing user interfaces for different screen resolutions? How does Android address this?

Ans:

Challenges include varying screen sizes, densities (ldpi-xxxhdpi), aspect ratios, and orientation changes. UI may appear stretched or misaligned if not handled properly. Android addresses this by:

- Using **dp/sp units** instead of pixels for scaling.
- Providing **multiple resource folders** (layout-large, drawable-hdpi etc.).
- Using **ConstraintLayout** for responsive design.
- Supporting **vector graphics** and **adaptive layouts**. This ensures apps look good on all devices.

Q7. Explain the lifecycle of an Android Activity.

Ans:

Activity lifecycle defines the states an activity transitions through:

- **onCreate()** → Initialize components and UI.
- **onStart()** → Activity becomes visible to user.
- **onResume()** → App ready for user interaction (foreground).
- **onPause()** → Partially hidden; save UI changes.
- **onStop()** → No longer visible; release resources.
- **onDestroy()** → Activity removed from memory.

Understanding lifecycle helps manage resources and prevent crashes.

Q8. Describe the use of Services in Android.

Ans:

A Service performs **background tasks** without UI such as:

- Playing music
- Uploading or downloading files
- Syncing data with server Types:
 1. **Started Service** – Works independently until stopped.
 2. **Bound Service** – Provides interaction with components when bound.
 3. **Foreground Service** – Shows notification and keeps service alive.

Services improve performance by offloading long tasks from UI thread.

Q9. What are Broadcast Receivers in Android? Give an example.

Ans:

Broadcast Receivers handle broadcasted system or application events. Example events:

- Battery low
- Network change
- Boot completed

Receivers register in the Manifest or at runtime. Example: An app detects when charging is connected and updates UI accordingly. They enable apps to react to global system changes.

Q10. What is a Content Provider in Android? Why is it used?

Ans:

A Content Provider manages and shares structured data between apps using **URI** (Uniform Resource Identifier). Uses:

- Share Contacts data
- Access images with MediaStore
- Provide custom data to other apps

It ensures data security, standardized access, and inter-application communication.

11. Describe the use of Services in Android.

Ans: - A Service is a component in Android that performs long-running operations in the background, without interacting with the user interface. Services are used for tasks such as:

- Playing music or videos
- Downloading files
- Handling network requests
- Performing database operations

Services can run even when the user is not interacting with the app, and they can be started and stopped as needed.

12. What are Broadcast Receivers in Android? Give an example.

Ans: - A Broadcast Receiver is a component in Android that listens for system-wide broadcast messages, such as:

- Battery low
- Screen on/off
- Network connectivity changes
- SMS received

When a broadcast is received, the Broadcast Receiver can perform an action, such as starting a Service or Activity.

Example: A Broadcast Receiver can be used to listen for the "BOOT_COMPLETED" broadcast, which is sent when the device boots up. The Broadcast Receiver can then start a Service to perform some initialization tasks.

13. What is a Content Provider in Android? Why is it used?

Ans: - A Content Provider is a component in Android that manages access to a shared repository of data, such as a database or file system. Content Providers provide a standardized interface for accessing and manipulating data, allowing different apps to share data with each other.

Content Providers are used for:

- Sharing data between apps
- Providing a centralized data repository
- Implementing data security and permissions

14. Explain the activity lifecycle in Android. Describe any five methods with its purpose and example.

Ans: - The Activity lifecycle is the series of methods that are called when an Activity is created, started, resumed, paused, stopped, or destroyed.

Here are five methods with their purpose and example:

1. **onCreate(Bundle savedInstanceState):** Called when the Activity is created.

- Purpose: Initialize the Activity's UI and variables.
- Example: Set up the layout and initialize variables in onCreate().

2. **onStart():** Called when the Activity becomes visible to the user.

- Purpose: Perform actions that should be done when the Activity is visible.
- Example: Start an animation or a video playback in onStart().

3. **onResume():** Called when the Activity gains focus and is ready to interact with the user.

- Purpose: Perform actions that require the Activity to be in the foreground.
- Example: Start a game or a music playback in onResume().

4. **onPause():** Called when the Activity loses focus and is no longer in the foreground.

- Purpose: Perform actions that should be done when the Activity is paused.
- Example: Pause a game or a music playback in onPause().

5. **onDestroy():** Called when the Activity is destroyed.

- Purpose: Clean up resources and perform final actions.
- Example: Release system resources and cancel any pending tasks in onDestroy().

15. Differentiate between implicit and explicit intents.

Ans:- Implicit Intent: An Intent that specifies the action to be performed, but not the component that should perform it.

Example: `Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse("https://www.example.com"));`

Explicit Intent: An Intent that specifies both the action and the component that should perform it.

Example: `Intent intent = new Intent(this, MyActivity.class);`

16. Define SharedPreferences in Android. Discuss its advantages and limitations

Ans:- Shared Preferences is a simple key-value store that allows apps to store small amounts of data, such as preferences or settings.

Advantages:

- Easy to use
- Lightweight
- Suitable for small amounts of data

Limitations:

- Not suitable for large amounts of data
- Not secure
- Data is stored in plain text

17. Explain the importance of data storage in mobile applications.

Ans:- Data storage is essential in mobile applications because:

- Mobile devices can lose data due to battery drain, app crashes, or device failure.
- Users expect their data to be persisted even when the app is closed or the device is restarted.
- Data storage allows apps to provide a seamless user experience and maintain user progress.

18. List and explain the types of data storage available in Android.

Ans:- Here are the types of data storage available in Android:

1. **SharedPreferences:** A simple key-value store for small amounts of data.
2. **Internal Storage:** A private storage area for app-specific data.
3. **External Storage:** A public storage area for data that can be accessed by other apps.
4. **SQLite Database:** A self-contained database that allows apps to store structured data.
5. **Content Providers:** A component that manages access to a shared repository of data.

19. What is Shared Preferences? Explain its advantages and limitations.

Ans:- Shared Preferences is a simple key-value store that allows apps to store small amounts of data, such as preferences or settings.

Advantages:

- Easy to use
- Lightweight
- Suitable for small amounts of data

Limitations:

- Not suitable for large amounts of data
- Not secure
- Data is stored in plain text

20. Describe the purpose of SQLite in mobile application development.

Ans:- SQLite is a self-contained database that allows apps to store structured data. SQLite is used in mobile application development for:

- Storing and managing large amounts of data
- Providing a robust and efficient data storage solution
- Supporting complex queries and data relationships

SQLite is a popular choice for mobile app development due to its ease of use, flexibility, and performance.

21. Compare synchronous and asynchronous network calls in Android with examples.

Ans:-

- Synchronous network calls: Block the main thread until the network request is complete.
 - Example: Using HttpURLConnection to make a network request on the main thread.
- Asynchronous network calls: Perform network requests in the background, allowing the main thread to continue executing.

Example: Using AsyncTask or Retrofit to make network requests asynchronously.

Asynchronous network calls are preferred in Android because they prevent the app from becoming unresponsive and reduce the risk of ANRs (Application Not Responding).

22. Explain the steps to play video in an Android application with playback controls.

Ans:-

1. Add a VideoView to the layout.
2. Set up the video source using setVideoURI() or setVideoPath().
3. Create a MediaController and set it to the VideoView.
4. Start the video playback using start().
5. Implement playback controls such as play, pause, and seek using the MediaController.

23. Explain the role of networking in mobile apps and why it is important.

Ans:-

Networking is essential in mobile apps for:

- Fetching data from remote servers
- Sending data to remote servers
- Real-time communication with servers
- Accessing cloud-based services

Networking is important because it allows mobile apps to:

- Provide a dynamic and interactive user experience
- Access remote data and services
- Enable real-time communication and updates

24. Explain the process of image loading and manipulation in Android.

Ans:-

1. Load the image from a resource, file, or URL.
2. Use an ImageView to display the image.
3. Use image processing libraries such as Glide or Picasso to load and manipulate images.
4. Apply transformations and effects to the image using libraries such as Android's built-in Bitmap class.

25. What are media playback controls, and how are they implemented?

Ans:-

Media playback controls are UI components that allow users to control media playback, such as play, pause, and seek.

Implementation:

1. Use a MediaController to provide playback controls for video and audio playback.
2. Create a custom UI component to provide playback controls.
3. Use the MediaPlayer class to control media playback.

26. Describe the role of Handlers in background processing.

Ans:-

Handlers are used to:

- Post messages and runnables to a thread's message queue.
- Handle messages and runnables posted to a thread's message queue.

Handlers are useful for:

- Updating the UI thread from a background thread.
- Scheduling tasks to run on a specific thread.

27. List any five multimedia components supported in Android.

Ans:-

1. MediaPlayer
2. VideoView
3. AudioManager
4. SoundPool
5. Camera

28. What is AsyncTask? Explain multithreading and background task management in Android.

Ans:- AsyncTask is a class that allows you to perform background tasks on a separate thread, while providing a simple way to update the UI thread.

Multithreading in Android:

- Allows apps to perform multiple tasks concurrently.
- Improves app responsiveness and performance.

Background task management:

- Use AsyncTask or HandlerThread to perform background tasks.
- Use Service or IntentService to perform long-running tasks.

29. Write Android program to load an image from drawable and apply zoom-in and zoom-out animation.

Ans:-

```
ImageView imageView = findViewById(R.id.imageView);
imageView.setImageResource(R.drawable.image);
imageView.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Animation zoomIn = AnimationUtils.loadAnimation(getApplicationContext(),
R.anim.zoom_in);
        imageView.startAnimation(zoomIn);
    }
});
// zoom_in.xml
<?xml version="1.0" encoding="utf-8"?>
<scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromXScale="1.0"
    android:fromYScale="1.0"
    android:pivotX="50%"
    android:pivotY="50%"
```

```
android:toXScale="2.0"  
android:toYScale="2.0" />
```

30. Explain the difference between GET and POST requests.

Ans:-

- GET request:

- Used to retrieve data from a server.
- Data is sent in the URL.
- Limited data capacity.

- POST request:

- Used to send data to a server.
- Data is sent in the request body.
- Can handle large amounts of data.

31. Discuss authentication vs authorization. Explain OAuth 2.0 workflow.

Ans:-

Authentication: Verifying the identity of a user or device.

Authorization: Granting access to resources or actions based on user identity or permissions.

OAuth 2.0 is an authorization framework that allows users to grant third-party apps access to their resources without sharing passwords.

OAuth 2.0 workflow:

1. Client registration: The client app registers with the authorization server.
2. Authorization request: The client app requests authorization from the user.
3. Authorization grant: The user grants authorization to the client app.
4. Token request: The client app requests an access token from the authorization server.
5. Token response and refresh token: The client app receives an access token and refresh token.
6. Protected resource access: The client app uses the access token to access protected resources.

32. What is the Retrofit library in Android, and why is it used?

Ans:-

Retrofit is a library for making HTTP requests in Android. It provides a simple and efficient way to interact with RESTful APIs.

Retrofit is used because it:

- Simplifies HTTP requests and responses.
- Supports synchronous and asynchronous requests.
- Provides built-in support for popular converters and adapters.

33. Explain how to access location-based services in Android.

Ans:-

1. Request location permissions in the AndroidManifest.xml file.
2. Use the LocationManager class to access location services.
3. Use the FusedLocationProviderClient class to get the current location or request location updates.

34. How can sensors be used in Android applications?

Ans:-

Sensors can be used to:

- Detect device orientation and movement.
- Measure environmental conditions such as light and temperature.
- Provide input for games and interactive applications.

Android provides a SensorManager class to access and manage sensors.

35. Distinguish between Transition Effects and Tween Animation in Android applications.

Ans:-

Transition Effects: Used to create animations between different states of a view or activity.

Tween Animation: Used to create animations that change the properties of a view over time.

Transition Effects are used for more complex animations, while Tween Animation is used for simpler animations.

36. Explain the importance of Test-Driven Development (TDD) in Android.

Ans:-

TDD is important in Android because it:

- Ensures that code is testable and meets requirements.
- Reduces bugs and errors.
- Improves code quality and maintainability.

37. What is performance optimization in Android applications?

Ans:-

Performance optimization in Android involves:

- Improving app responsiveness and speed.
- Reducing battery drain and resource usage.
- Optimizing code and layouts for better performance.

38. Explain GPS, maps integration and sensor handling in Android

Ans:-

- GPS: Used to get the device's current location.
- Maps integration: Used to display maps and provide location-based services.
- Sensor handling: Used to access and manage device sensors such as GPS, accelerometer, and gyroscope.

39. Explain RESTful APIs, JSON parsing, and authentication (OAuth and JWT) in Android app development.

Ans:-

RESTful APIs: Used to interact with web services and retrieve data.

JSON parsing: Used to parse JSON data from web services.

Authentication: Used to secure API requests and authenticate users.

OAuth: An authorization framework that allows users to grant access to their resources.

JWT: A token-based authentication mechanism that securely transmits information between parties.

40. Discuss GPS, maps integration, and sensor data processing in Android.

Ans:-

- GPS: Used to get the device's current location and provide location-based services.
- Maps integration: Used to display maps and provide location-based services.
- Sensor data processing: Used to access and process data from device sensors such as GPS, accelerometer, and gyroscope.

41. Explain unit testing, debugging techniques, and memory management in Android.

Ans:-

Unit testing: Unit testing involves testing individual components or units of code to ensure they function as expected. In Android, JUnit and Mockito are popular testing frameworks.

Debugging techniques: Debugging involves identifying and fixing errors in the code. Android Studio provides a built-in debugger, and techniques like Logcat, breakpoint debugging, and Analyze Stacktrace are useful for debugging.

Memory management: Memory management is crucial in Android to prevent memory leaks and optimize performance. Techniques like using weak references, avoiding static variables, and using memory profiling tools can help manage memory effectively.

42. Explain the steps to play video in an Android application with playback controls.

Ans:-

1. Add a VideoView to the layout.
2. Set up the video source using setVideoURI() or setVideoPath().
3. Create a MediaController and set it to the VideoView.
4. Start the video playback using start().

5. Implement playback controls such as play, pause, and seek using the MediaController.

Example code:

```
VideoView videoView = findViewById(R.id.videoView);  
  
videoView.setVideoURI(Uri.parse("android.resource://" + getPackageName() + "/" +  
R.raw.video));  
  
MediaController mediaController = new MediaController(this);  
  
videoView.setMediaController(mediaController);  
  
videoView.start();
```

43. Discuss the importance of background processing and multithreading in Android with examples.

Ans:-

Background processing and multithreading are essential in Android to:

- Perform long-running tasks without blocking the main thread.
- Improve app responsiveness and performance.
- Take advantage of multi-core processors.

Examples:

- Using AsyncTask to perform background tasks, such as downloading data or performing database operations.
- Using Thread or HandlerThread to perform long-running tasks.
- Using RxJava or Kotlin Coroutines to manage asynchronous tasks and improve code readability.

44. Explain the role of Camera and Photo Capture in mobile applications with an example.

Ans:-

Camera and photo capture are essential features in many mobile applications, such as:

- Social media apps that allow users to share photos.
- Camera apps that provide advanced camera features.
- Photo editing apps that allow users to edit and enhance photos.

To use the camera in an Android app:

1. Request camera permissions in the AndroidManifest.xml file.
2. Use the Camera or Camera2 API to access the camera.
3. Use an Intent to capture a photo or video.

Example code:

```
Intent intent = new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
startActivityForResult(intent, REQUEST_IMAGE_CAPTURE);

In onActivityResult():

if (requestCode == REQUEST_IMAGE_CAPTURE && resultCode == RESULT_OK) {
    Bundle extras = data.getExtras();
    Bitmap imageBitmap = (Bitmap) extras.get("data");
    ImageView imageView = findViewById(R.id.imageView);
    imageView.setImageBitmap(imageBitmap);
}
```